MALLA REDDY ENGINEERING COLLEGE FOR WOMEN

(Autonomous Institution-UGC, Govt. of India)
Accredited by NAAC with 'A+' Grade | Programmes Accredited by NBA
NIRF Institutional Ranking 2025 – Rankband(201-300), MHRD, Govt. of India
Approved by AICTE, Affiliated to JNTUH, ISO 9001:2015 Certified Institution
Maisammaguda, Dhulapally, Secunderabad 500100.

DEPARTMENT: H&S **SECTION:AIML-A,B**	**COURSE**: I B.TECH I SEM	**SUBJECT**: Programming for Problem Solving (2605ES01)
Unit-1, 2 & 3 (Part)	**R26** REGULATION	**ACADEMIC YEAR:** 2026-2027

Course Project-1

CATEGORY A: MANAGEMENT & INFORMATION SYSTEMS

No.	Course Project	Modules	Major Concepts	CO	PO	PSO	SDG
1	**Hospital Patient Management and Billing System**	1. Patient Registration 2. Doctor & Department Allocation 3. Appointment Scheduling 4. Billing & Payment 5. Search & Sorted Reports	Unit I: menu with switch, if-else, do-while, bill arithmetic Unit II: top-down modules, math.h, pointer output parameters for totals Unit III: patient record arrays, linear search by ID, bubble sort by bill, enum for departments	CO 1–6	PO1, PO2, PO3, PO5	PSO1, PSO2	SDG 3
2	**University Result Processing and Rank Generation System**	1. Student & Marks Entry 2. Grade & CGPA Calculation 3. Rank Generation (Sorting) 4. Result Search by Roll No. 5. Result Analysis Report	Unit I: grade conditions, nested loops, GPA/percentage computation Unit II: functions for grade/CGPA, pointers returning topper and average Unit III: 2-D marks matrix, selection sort for ranking, binary search by roll no.	CO 1–6	PO1, PO2, PO3, PO5	PSO1, PSO2	SDG 4
3	**Banking and ATM Management System with Transaction History**	1. Account Creation 2. Deposit & Withdrawal 3. Fund Transfer 4. Transaction History 5. Account Search & Balance Report	Unit I: do-while menu, nested if, switch, balance arithmetic Unit II: modular deposit/withdraw/transfer, pointers for updated balance Unit III: account arrays, transaction log array, search account, sort by balance, enum transaction type	CO 1–6	PO1, PO2, PO3, PO5	PSO1, PSO2	SDG 8
4	**Library Book Issue, Return and Fine Management System**	1. Book Catalog 2. Member Registration 3. Issue & Return 4. Fine Calculation	Unit I: conditional loops, switch menu, fine slabs with if-else Unit II: book/member modules, pointers for fine and due-date output	CO 1–6	PO1, PO2, PO3, PO5	PSO1, PSO2	SDG 4

		5. Search & Sorted Reports	Unit III: book catalog arrays, linear & binary search, insertion sort by title/ID, enum book status				
5	**Employee Payroll and Tax Management System**	1. Employee Records 2. Salary Computation 3. Tax & Deductions 4. Payslip Generation 5. Salary Ranking & Search	Unit I: salary slabs, nested if, loops over employees Unit II: payroll modules, math.h, output parameters for gross/net/tax Unit III: employee arrays, selection sort by salary, search by ID, 2-D monthly pay matrix, enum designation	CO 1–6	PO1, PO2, PO3, PO5	PSO1, PSO2	SDG 8
6	**Online Shopping Cart and Inventory Control System**	1. Product Management 2. Cart Operations 3. Discount & Billing 4. Stock Update 5. Sales Report & Search	Unit I: menu loop, discount conditions, bill arithmetic Unit II: cart/inventory functions, pointers for stock and total update Unit III: product arrays, binary search by code, bubble sort by price, 2-D sales table, enum category	CO 1–6	PO1, PO2, PO3, PO5	PSO1, PSO2	SDG 12
7	**Hotel Room Reservation and Billing System**	1. Room Setup 2. Availability Search 3. Booking & Check-in 4. Check-out & Billing 5. Occupancy Report	Unit I: switch-based menu, tariff conditions, do-while Unit II: booking/checkout modules, pointers for bill and availability Unit III: 2-D floor×room matrix, linear search for free rooms, sort by tariff, enum room type	CO 1–6	PO1, PO2, PO3, PO5	PSO1, PSO2	SDG 8
8	**Railway Ticket Reservation and Seat Allocation System**	1. Train & Coach Setup 2. Seat Availability 3. Ticket Booking 4. Cancellation & Waiting List 5. Fare Calculation	Unit I: nested if, fare calculation, repetition until exit Unit II: reservation/cancellation functions, pointers for seats and fare Unit III: 2-D coach×seat array, search seat/PNR, sort waiting list, enum class (SL/3A/2A)	CO 1–6	PO1, PO2, PO3, PO5	PSO1, PSO2	SDG 9

CATEGORY B: SIMULATION, GAMES & ENGINEERING APPLICATIONS

No.	Course Project	Modules	Major Concepts	CO	PO	PSO	SDG
9	**Smart Multi-Level Parking Management System**	1. Slot Setup 2. Vehicle Entry & Slot Allocation 3. Vehicle Exit & Fee Calculation 4. Free Slot Search 5. Occupancy & Revenue Report	Unit I: entry/exit conditions, fee slabs, loops Unit II: fee and slot functions, math.h, pointers for slot and fee output Unit III: 2-D level×slot array, search free slot, sort by duration, enum vehicle type	CO 1–6	PO1, PO2, PO3, PO5	PSO1, PSO2	SDG 11
10	**Connect-Four Game with Win Detection and Scoreboard**	1. Board Initialization & Display 2. Move Validation 3. Win/Draw Detection 4. Score Management 5. Scoreboard Ranking	Unit I: turn-based loops, if-else win checks, do-while replay Unit II: move/validate/check-win functions, pointers for scores Unit III: 2-D game board, nested traversal for winning lines, sorted scoreboard array, enum player/cell state	CO 1–6	PO1, PO2, PO3, PO5	PSO1, PSO2	SDG 9
11	**Sudoku Solver and Validator**	1. Puzzle Input 2. Row/Column/Box Validation 3. Solver using Loops 4. Hint Generator 5. Completion Check	Unit I: nested loops, conditional checks, counters Unit II: validate/solve modules, pointer-based row/col/box checks Unit III: 9×9 2-D array, searching empty cells, sorting/duplicate check per row, enum difficulty	CO 1–6	PO1, PO2, PO3, PO5	PSO1, PSO2	SDG 4
12	**Matrix-Based Image Processing Simulator**	1. Pixel Matrix Input 2. Brightness & Contrast 3. Rotate, Flip & Transpose 4. Histogram & Statistics 5. Image Display	Unit I: nested loops, menu switch, intensity arithmetic Unit II: filter/rotate/flip functions, math.h, pointers for image stats Unit III: 2-D pixel matrix, transpose, min/max search, sorted histogram array, enum filter type	CO 1–6	PO1, PO2, PO3, PO5	PSO1, PSO2	SDG 9
13	**Elevator Scheduling Simulator**	1. Request Input 2. Nearest-Floor Scheduler 3. Direction Control 4. Movement Simulation 5. Trip Summary	Unit I: loop-driven simulation, direction conditions, switch Unit II: scheduler modules, pointers for current floor and distance Unit III: request array, selection sort by nearest floor, search pending requests, enum direction/state	CO 1–6	PO1, PO2, PO3, PO5	PSO1, PSO2	SDG 9

14	**Traffic Signal and Vehicle Flow Simulation**	1. Junction Setup 2. Signal Controller 3. Vehicle Arrival Simulation 4. Density Analysis 5. Waiting-Time Report	Unit I: timing loops, nested if, counters Unit II: signal controller functions, pointers for waiting-time output Unit III: 2-D junction×lane array, max/min search, sort lanes by density, enum signal state	CO 1–6	PO1, PO2, PO3, PO5	PSO1, PSO2	SDG 11
15	**Scientific and Statistical Calculator with Matrix Support**	1. Menu Driver 2. Arithmetic & Trigonometric Functions 3. Statistics (Mean, Median, SD) 4. Matrix Operations 5. Equation Solver	Unit I: menu switch, do-while, arithmetic expressions Unit II: math.h functions, pointers for mean/variance/median outputs Unit III: data arrays, sorting for median, matrix add/multiply/transpose/determinant, binary search	CO 1–6	PO1, PO2, PO3, PO5	PSO1, PSO2	SDG 9
16	**Weather Data Analysis and Forecast Tool**	1. Weather Data Entry 2. Statistical Analysis 3. Sorting & Median 4. Search by Day/Temperature 5. Season-wise Report	Unit I: condition ranges, loops over days, averages Unit II: statistics functions, pointers for max/min/avg output Unit III: monthly temperature 2-D array, bubble sort, binary search, enum season	CO 1–6	PO1, PO2, PO3, PO5	PSO1, PSO2	SDG 13

CATEGORY C: DATA ANALYSIS & PLANNING UTILITIES

No.	Course Project	Modules	Major Concepts	CO	PO	PSO	SDG
17	**Sports Tournament Management and Points Table Generator**	1. Team Registration 2. Match Result Entry 3. Points Update 4. Points Table Ranking 5. Team Search & Report	Unit I: match result conditions, nested loops, switch Unit II: points/net-run-rate functions, pointers for updated team stats Unit III: team arrays, 2-D results matrix, sort by points, search team, enum result type	CO 1–6	PO1, PO2, PO3, PO5	PSO1, PSO2	SDG 3
18	**Election Voting and Result Tabulation System**	1. Candidate Registration 2. Voter Validation & Voting 3. Vote Counting 4. Winner & Margin 5. Result Ranking	Unit I: do-while voting loop, validation conditions, counters Unit II: vote count/winner modules, pointers for winner and margin Unit III: constituency×candidate 2-D array, sort rankings, search candidate, enum party	CO 1–6	PO1, PO2, PO3, PO5	PSO1, PSO2	SDG 16

19	**Examination Seating Arrangement and Invigilation Allocation System**	1. Room & Bench Setup 2. Student Registration 3. Seat Allocation 4. Seat Search 5. Invigilator Allocation	Unit I: nested loops, branch/room conditions, counters Unit II: allocation functions, pointers for seats left and clashes Unit III: 2-D room×bench array, search student, sort by roll no., enum branch	CO 1–6	PO1, PO2, PO3, PO5	PSO1, PSO2	SDG 4
20	**Personal Finance and Expense Tracker with Monthly Analysis**	1. Income Entry 2. Expense Entry 3. Budget Check 4. Monthly Analysis 5. Top-Expense Report	Unit I: switch menu, budget conditions, loops Unit II: expense/savings functions, math.h, pointers for totals Unit III: 2-D month×category array, sort top expenses, search month, enum category	CO 1–6	PO1, PO2, PO3, PO5	PSO1, PSO2	SDG 1
21	**Course Timetable Generator and Clash Checker**	1. Subject & Faculty Input 2. Slot Allocation 3. Clash Detection 4. Free Slot Search 5. Timetable Display	Unit I: nested loops, clash conditions, menu loop Unit II: slot allocation/verification functions, pointers for status Unit III: 2-D day×period array, search free slot, sort by subject load, enum days	CO 1–6	PO1, PO2, PO3, PO5	PSO1, PSO2	SDG 4
22	**Loan EMI Amortization and Comparison Planner**	1. Loan Input 2. EMI Calculation 3. Amortization Schedule 4. Plan Comparison 5. Plan Ranking & Search	Unit I: EMI arithmetic, loops over tenure, comparison conditions Unit II: EMI/interest functions, math.h pow(), pointers for totals Unit III: schedule arrays, 2-D plan table, sort plans by interest, binary search by tenure, enum loan type	CO 1–6	PO1, PO2, PO3, PO5	PSO1, PSO2	SDG 8

Course Project Instructions

- Each team should select/receive **one project title**.
- The project should be developed using **C**.
- Students must use the concepts discussed in **all of Units I, II and Arrays (first half of Unit III) in every project**.
- Each team should have **3 members**.if any team members has 4 divide it to two
- Students should prepare the source code, project report, test cases and output screenshots.
- Each team must give a project demonstration and viva.

Signature of the Faculty

Signature of the HOD